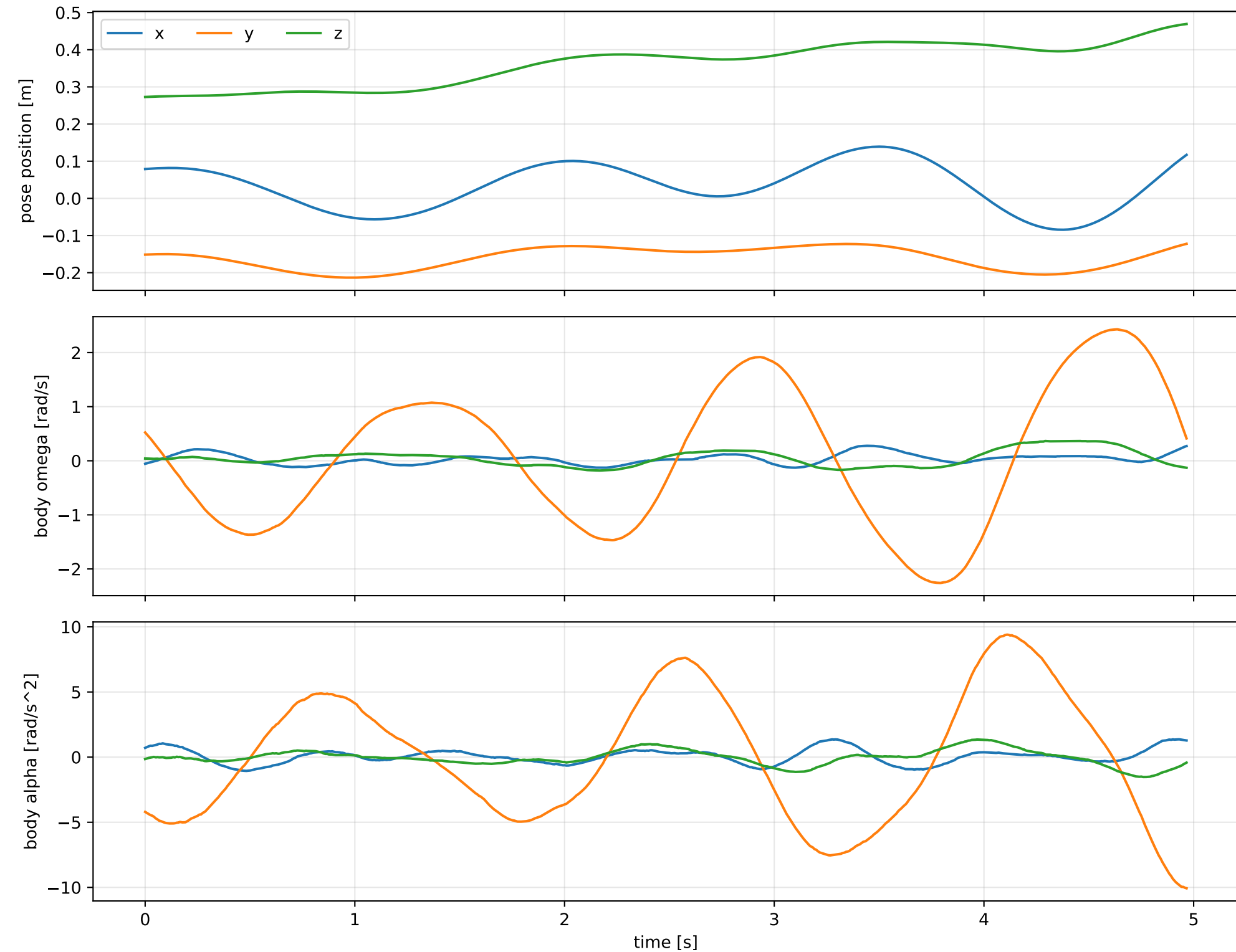
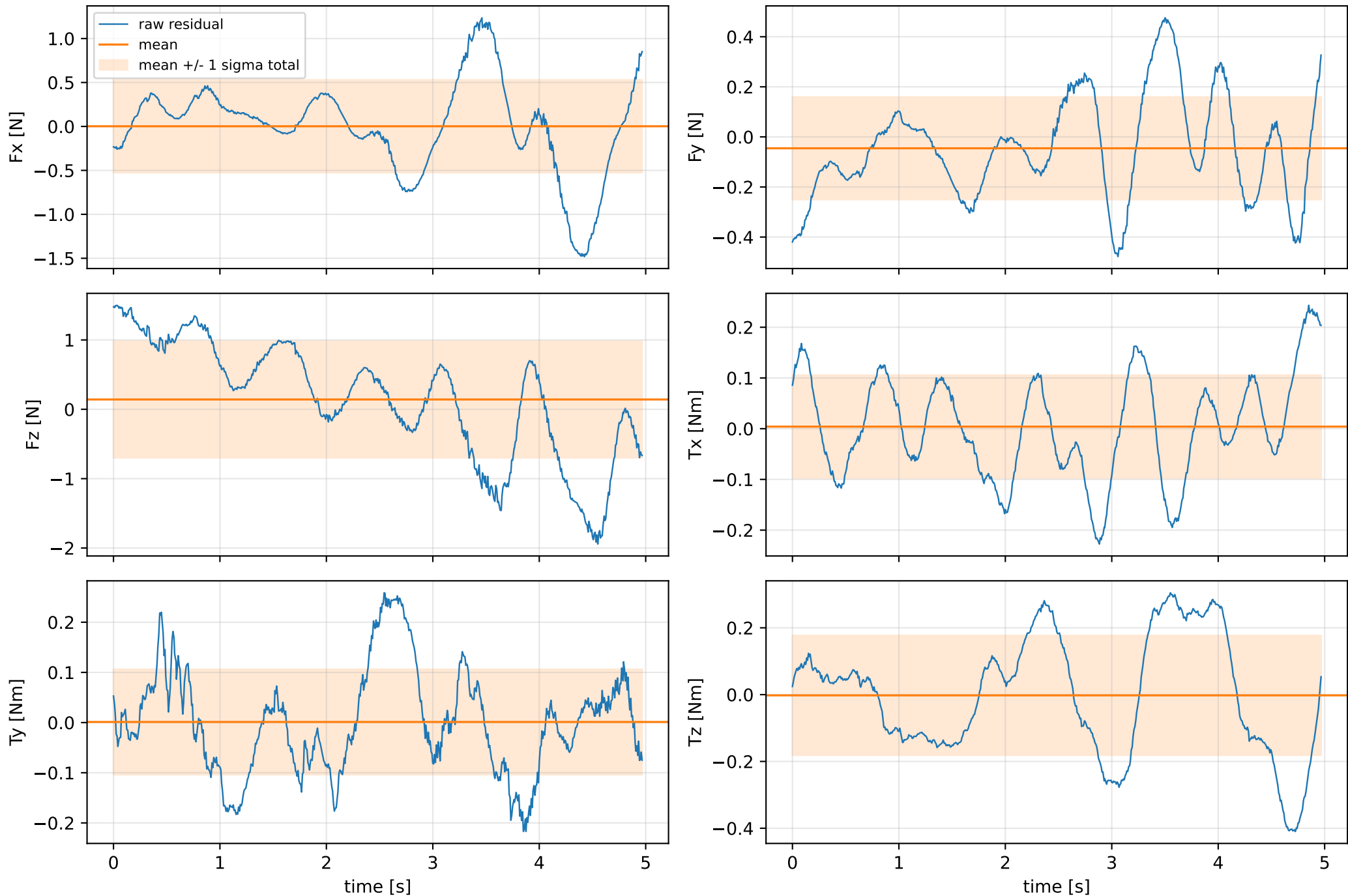


default: SG trajectory and derived motion



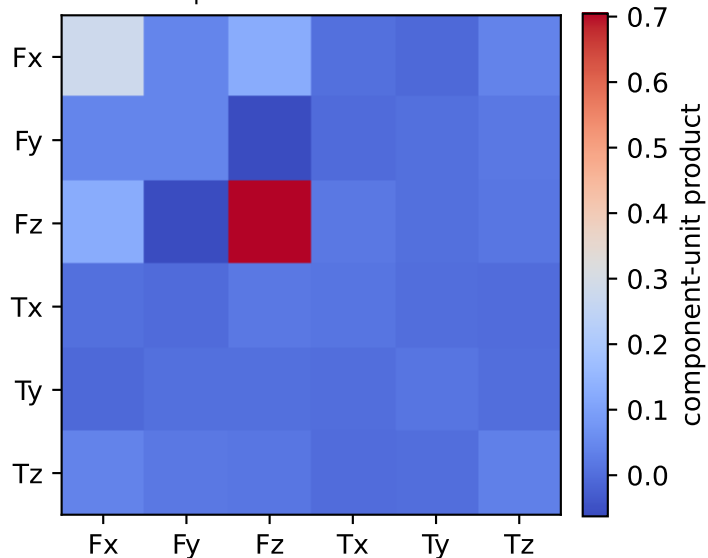
default: raw Newton-Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



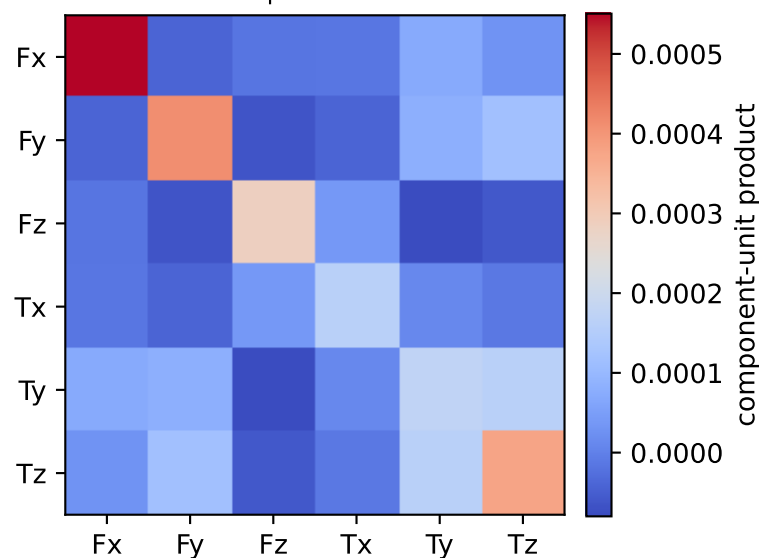
vector RMS about zero: force 1.0219 N, torque 0.2308 Nm; component std about mean: force [0.5268 0.2047 0.8393], torque [0.1014 0.1049 0.179]

default: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.266 0.741]

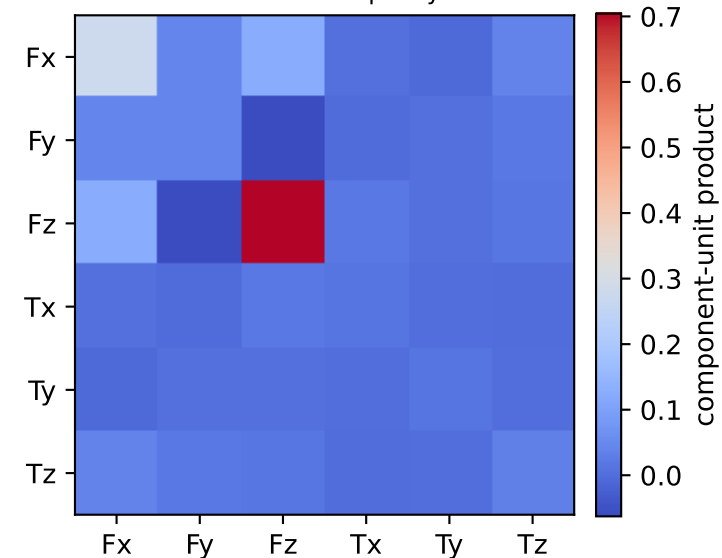
empirical total covariance



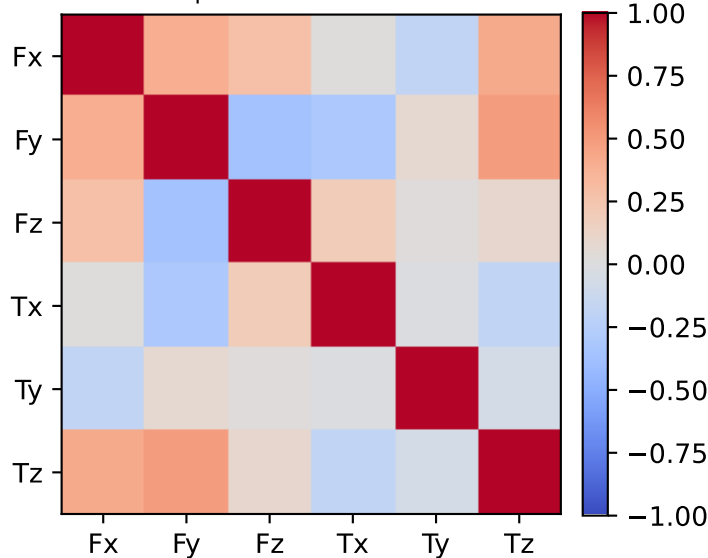
mean full-SG prediction covariance



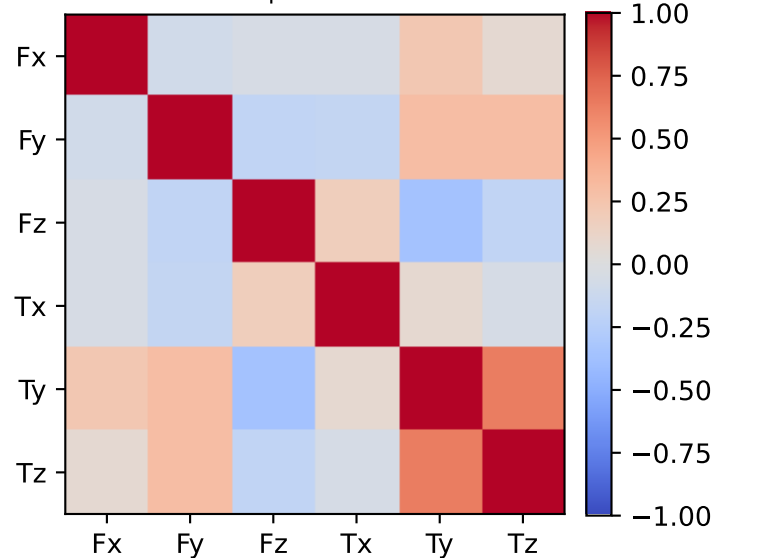
PSD excess model discrepancy covariance



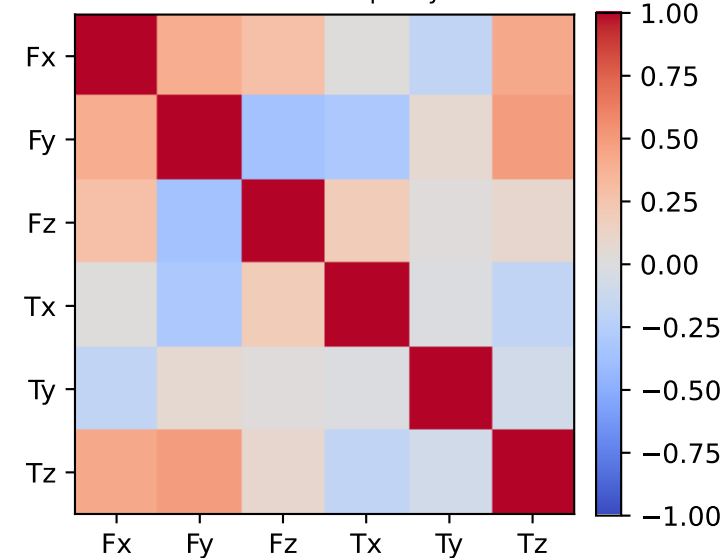
empirical total correlation



mean full-SG prediction correlation

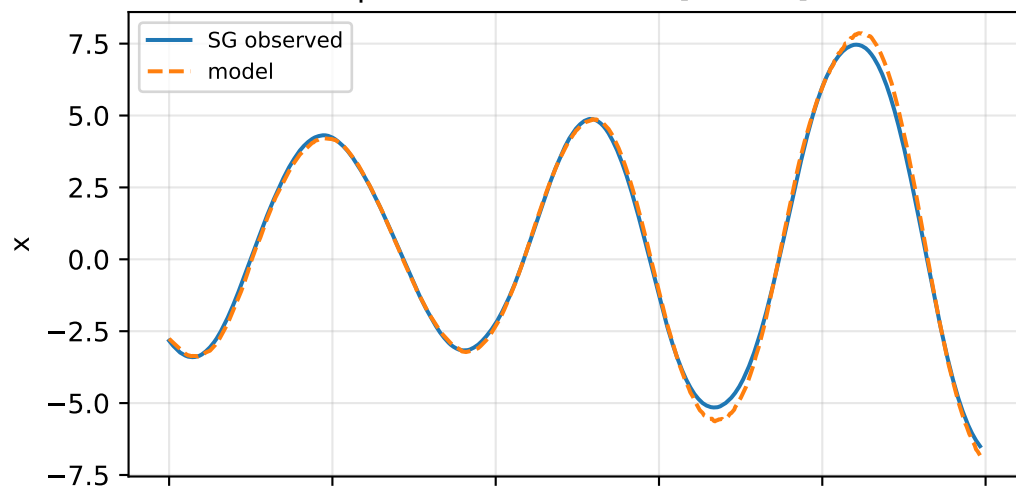


PSD excess model discrepancy correlation

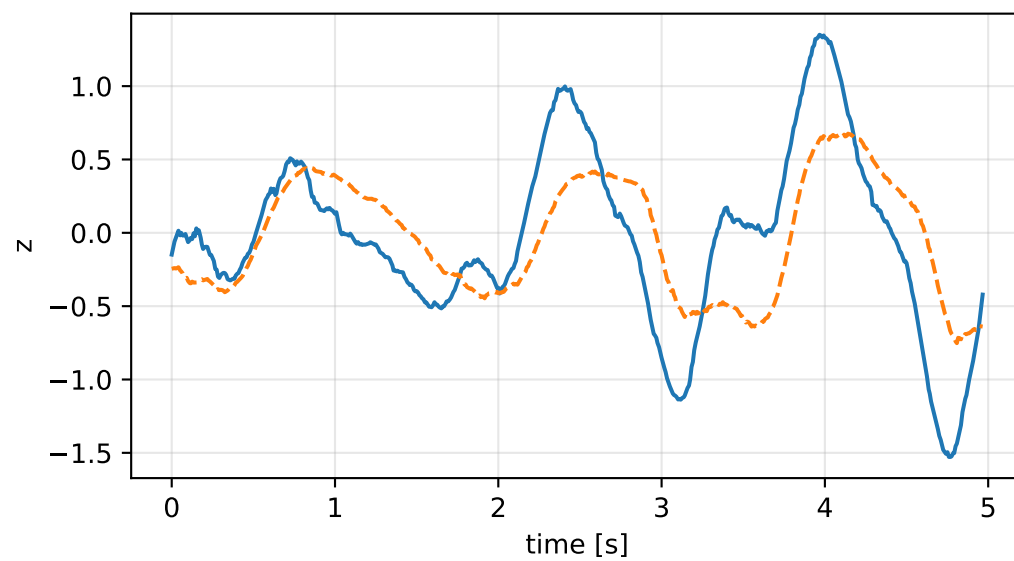
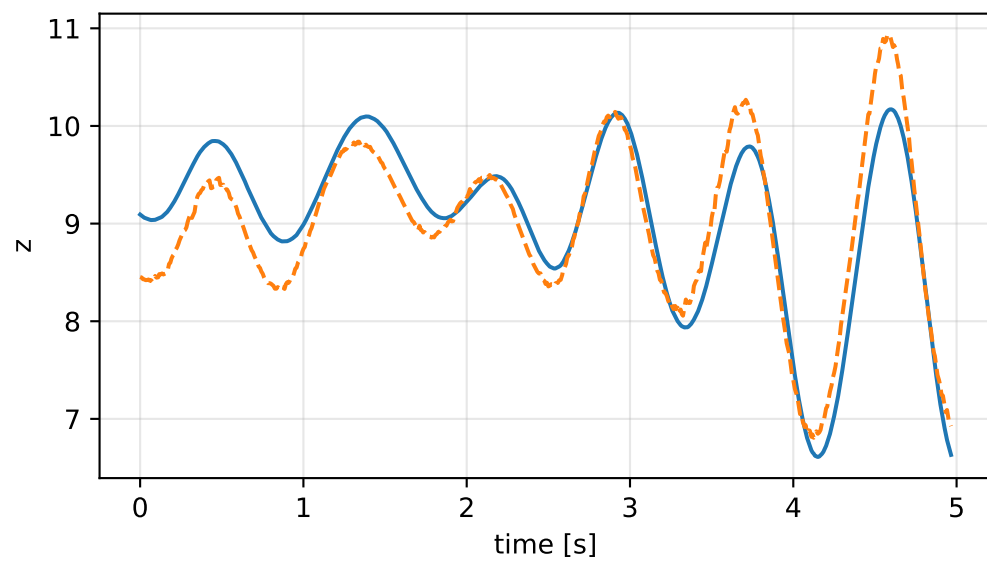
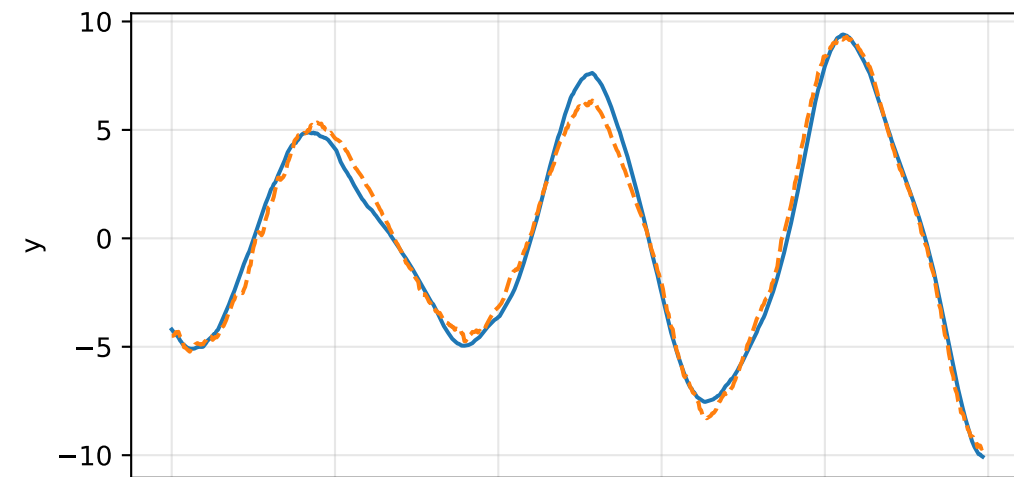
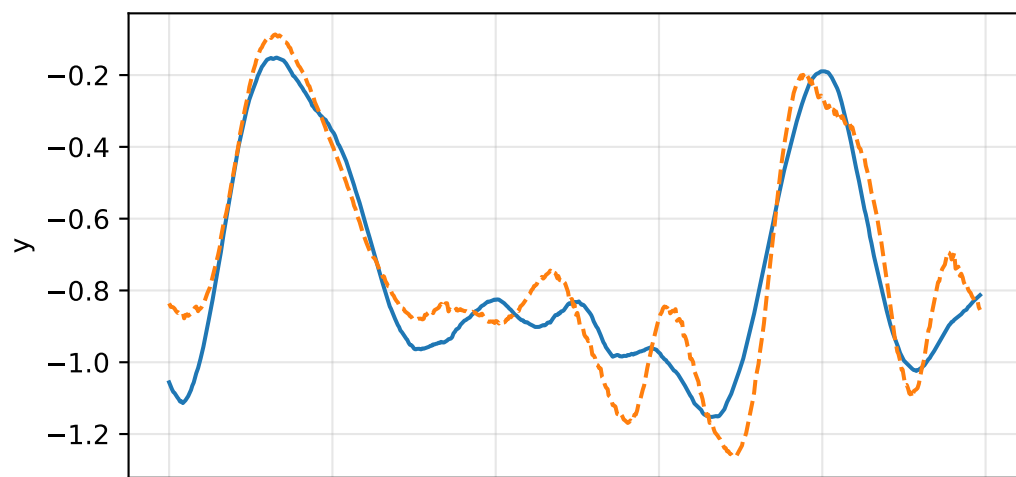
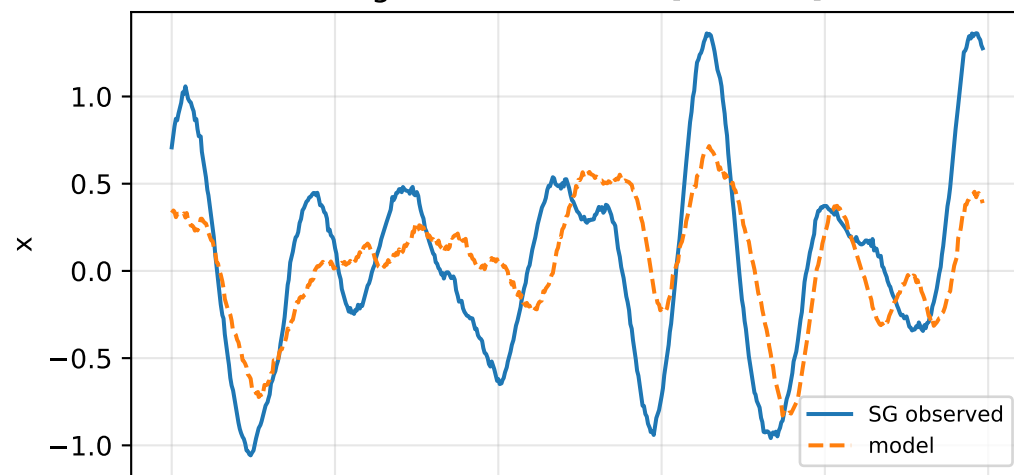


default: acceleration residual objective

specific acceleration [m/s²]

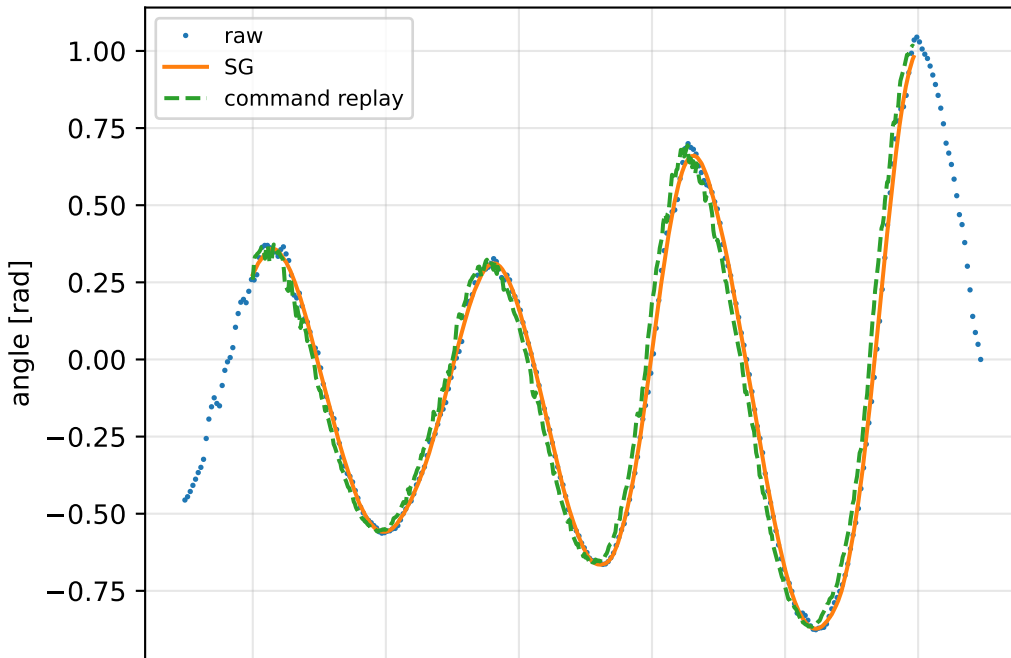


angular acceleration [rad/s²]

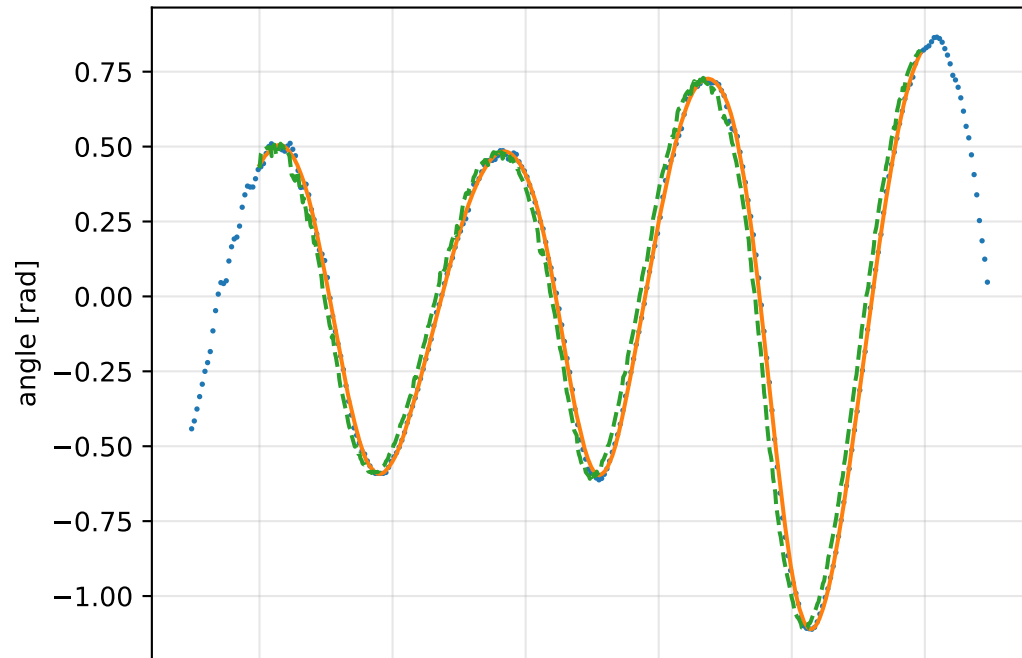


default: gimbal measurement audit (objective=measured_sg)

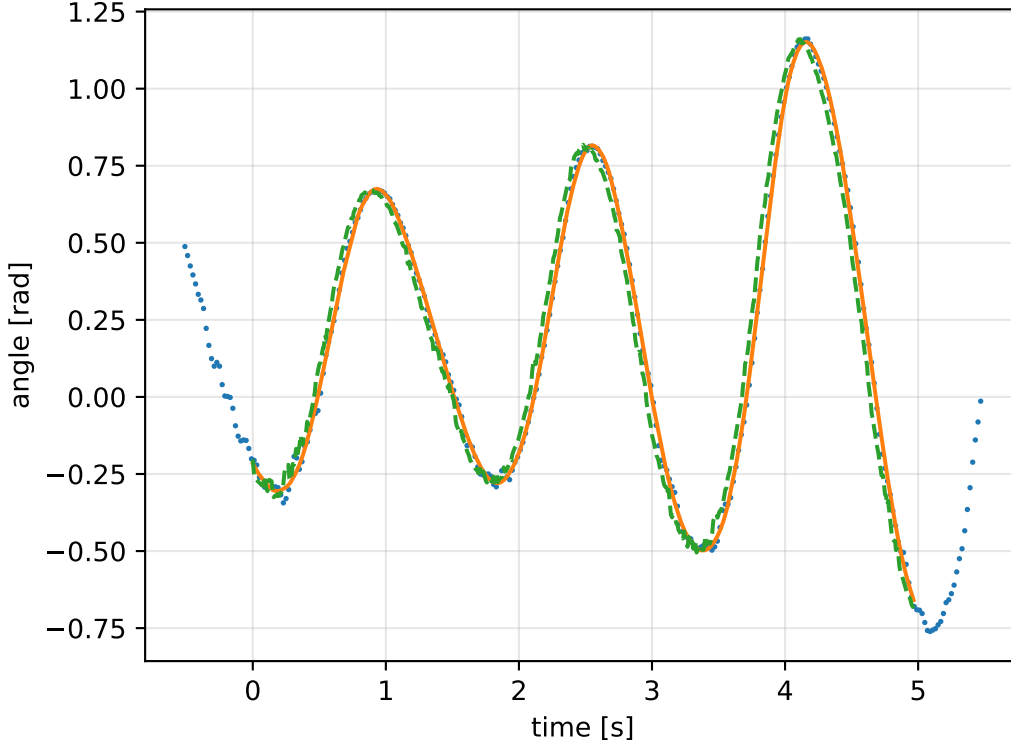
gimbal 1: RMSE=0.0759, max=0.2095 rad



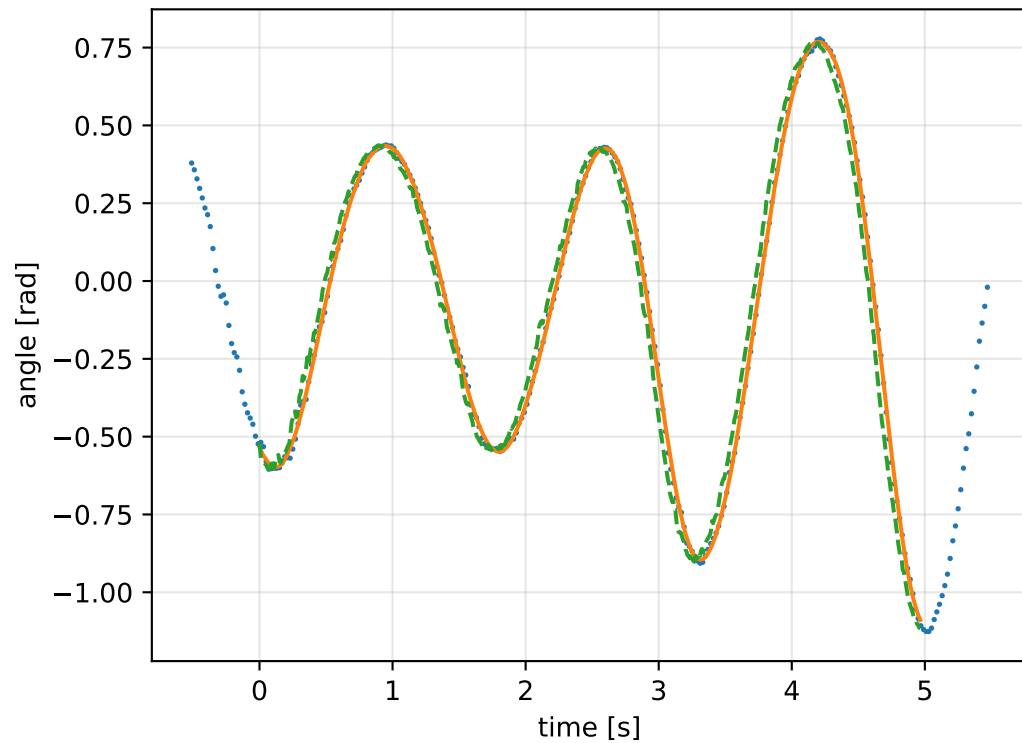
gimbal 2: RMSE=0.08099, max=0.2013 rad



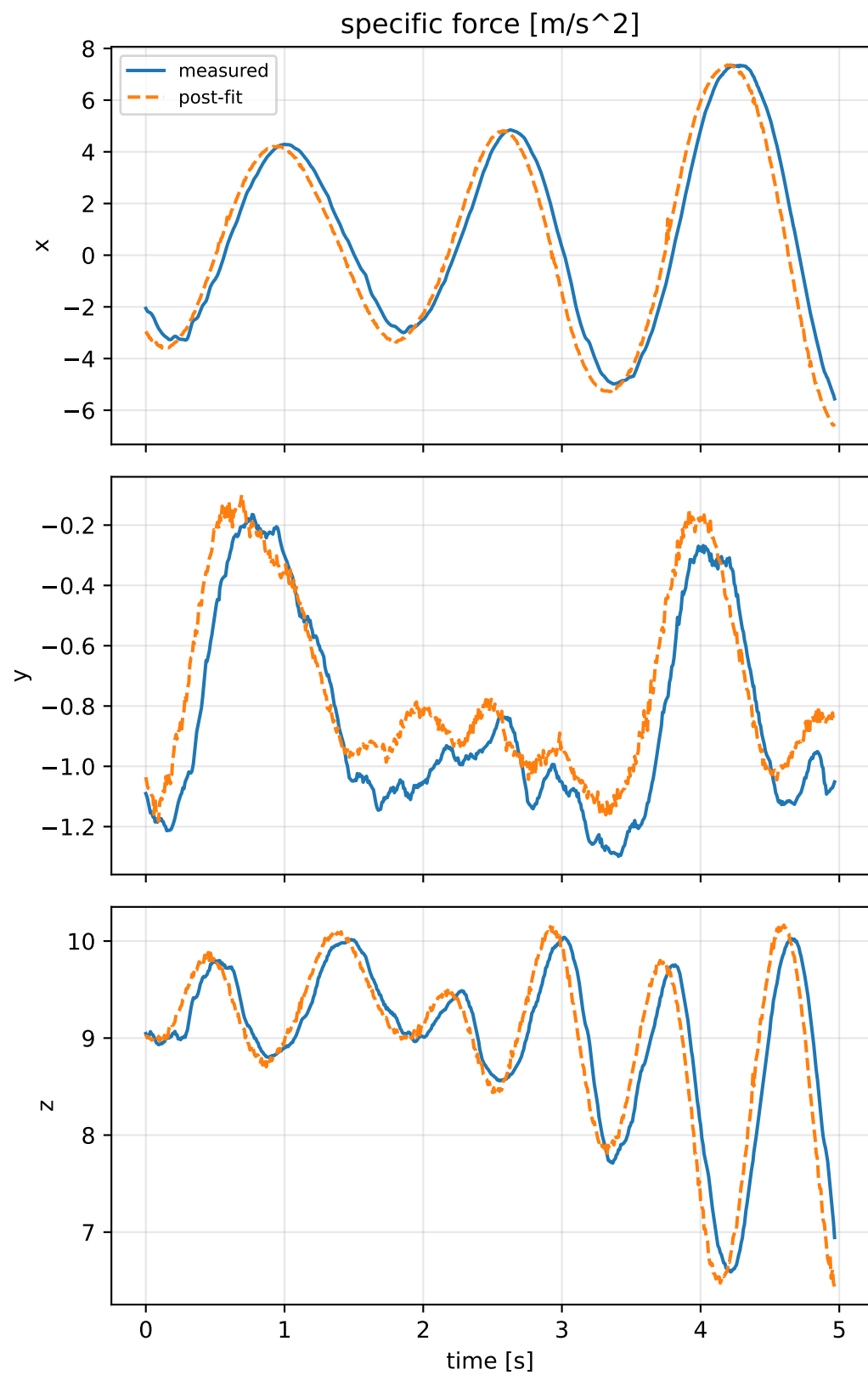
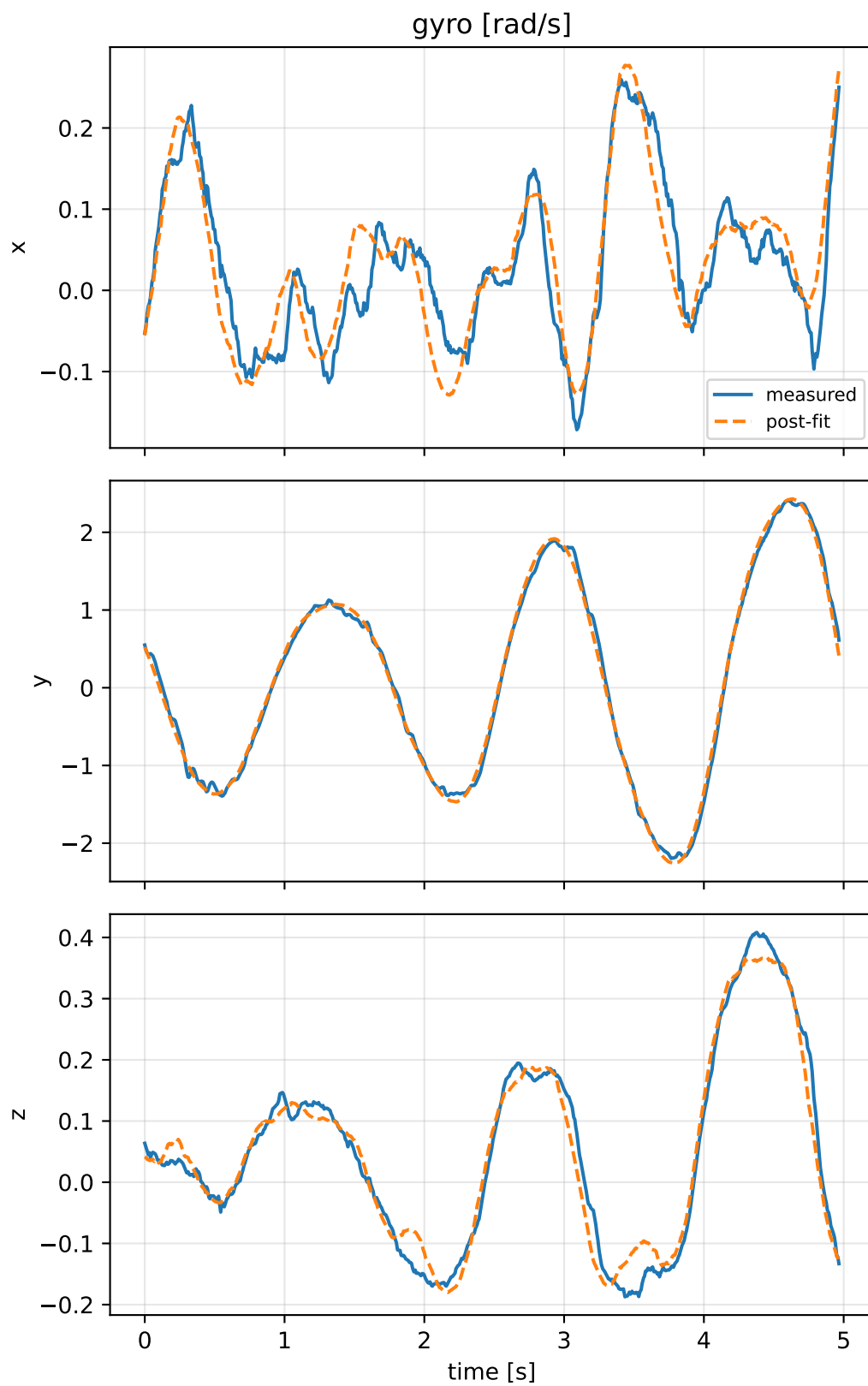
gimbal 3: RMSE=0.07393, max=0.172 rad



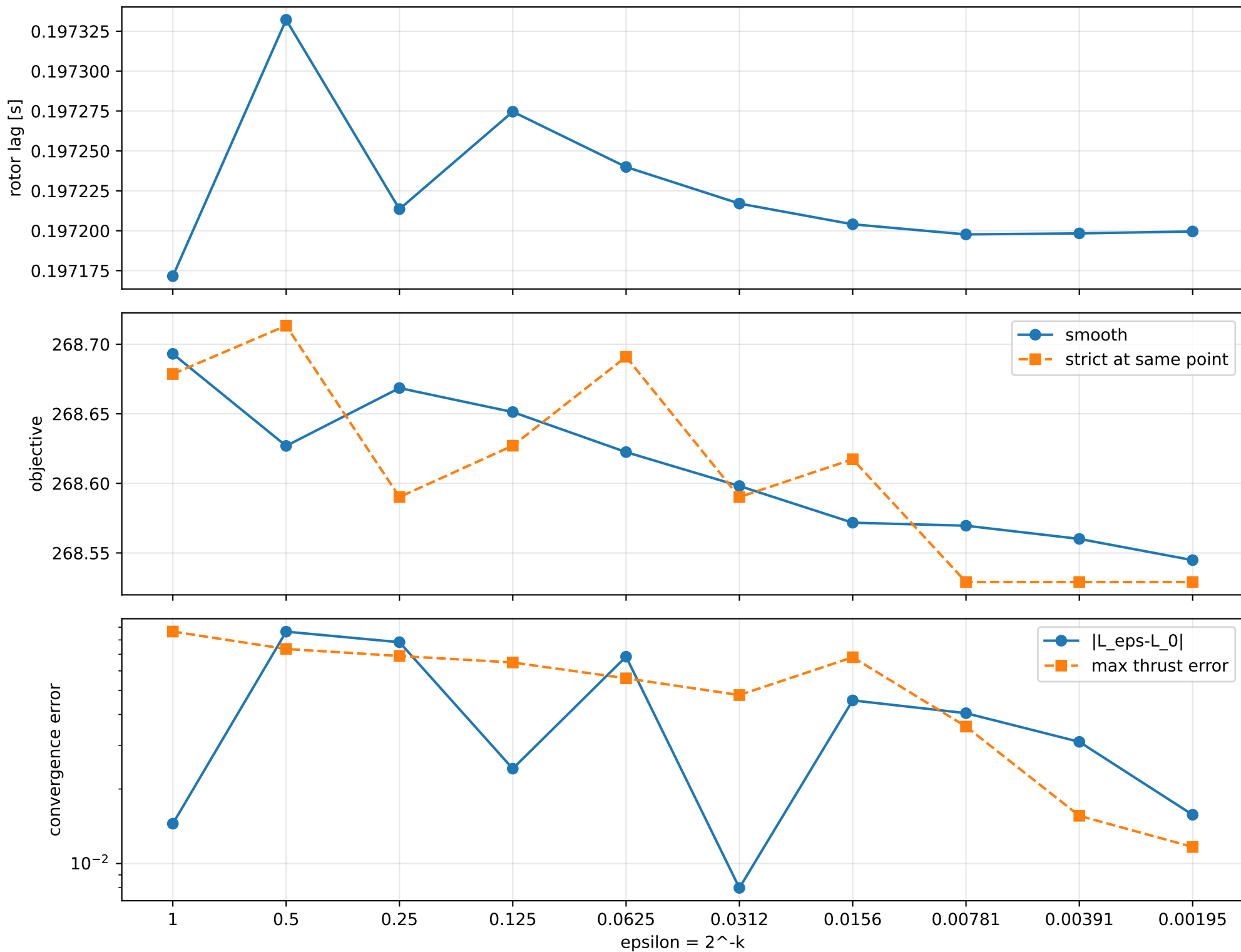
gimbal 4: RMSE=0.07131, max=0.1737 rad



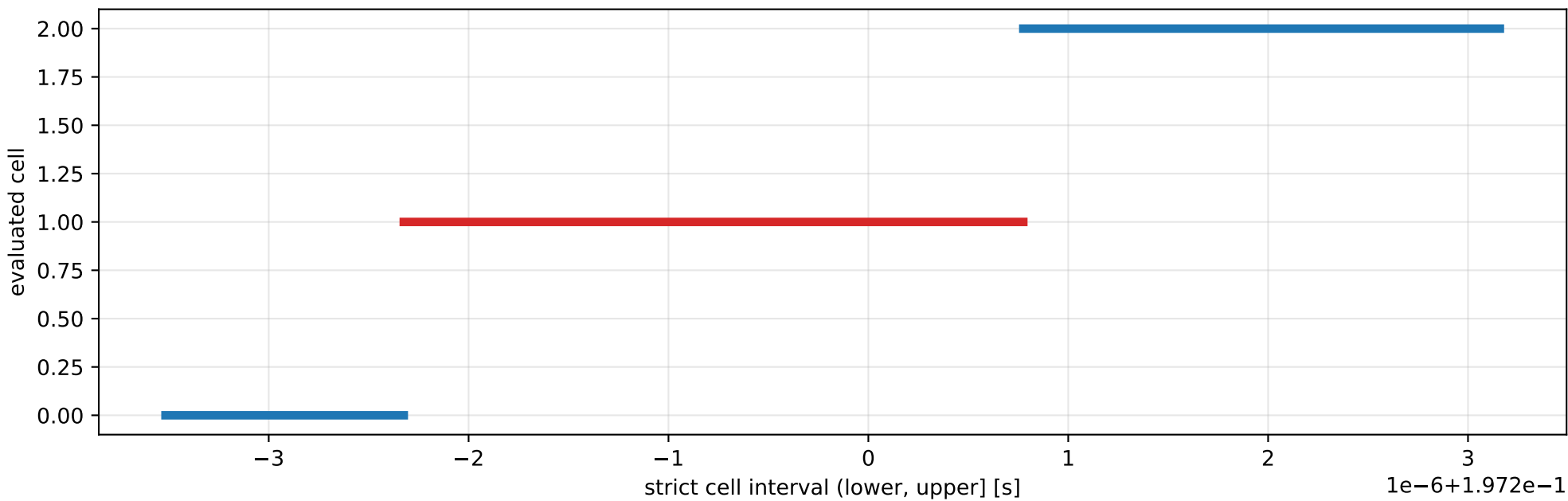
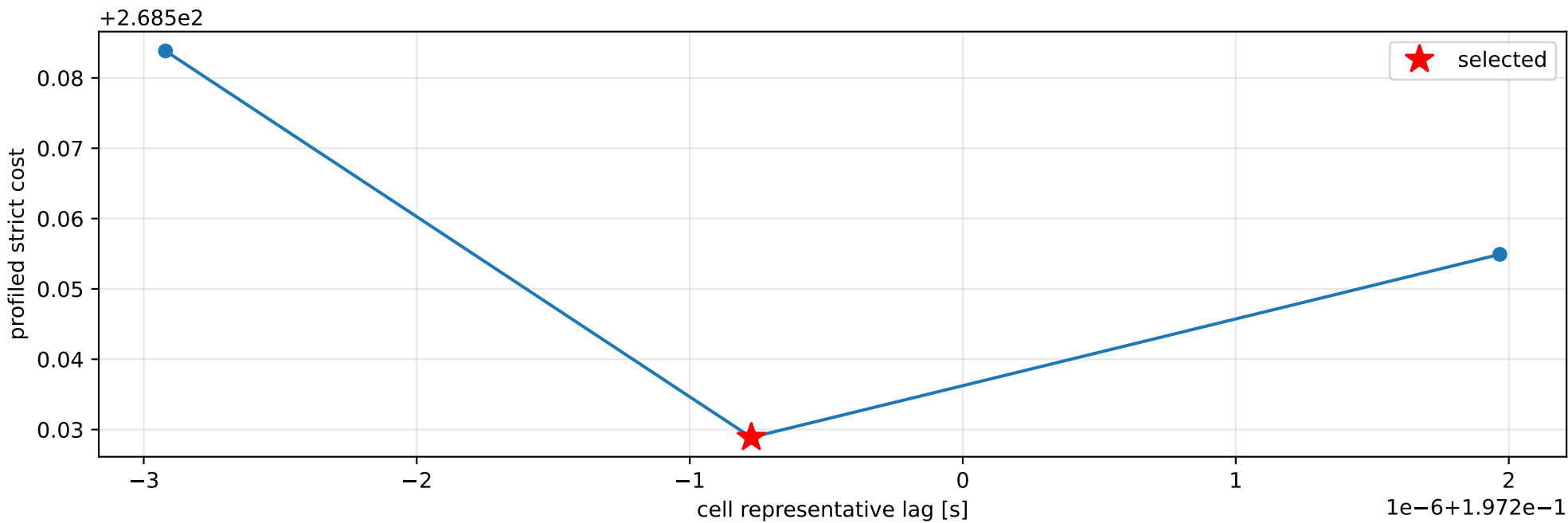
default: IMU comparison (diagnostic only)



default: smooth-to-strict continuation



default: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199564 s; data boundary=False

default: parameter estimate

Case: default

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.47573919

inertia [kg m²]:

[[3.53067464e-01 -4.27100120e-04 -5.73203733e-02]

[-4.27100120e-04 2.68867277e-01 3.30980041e-05]

[-5.73203733e-02 3.30980041e-05 5.96255973e-01]]

CoG body [m]: [0.01762808 -0.00411735 -0.04437562]

force effectiveness: [1.16405654 1.49895804 1.30895698 1.08525949]

scale-free J/m [m²]:

[[1.01580540e-01 -1.22880371e-04 -1.64915634e-02]

[-1.22880371e-04 7.73554236e-02 9.52257990e-06]

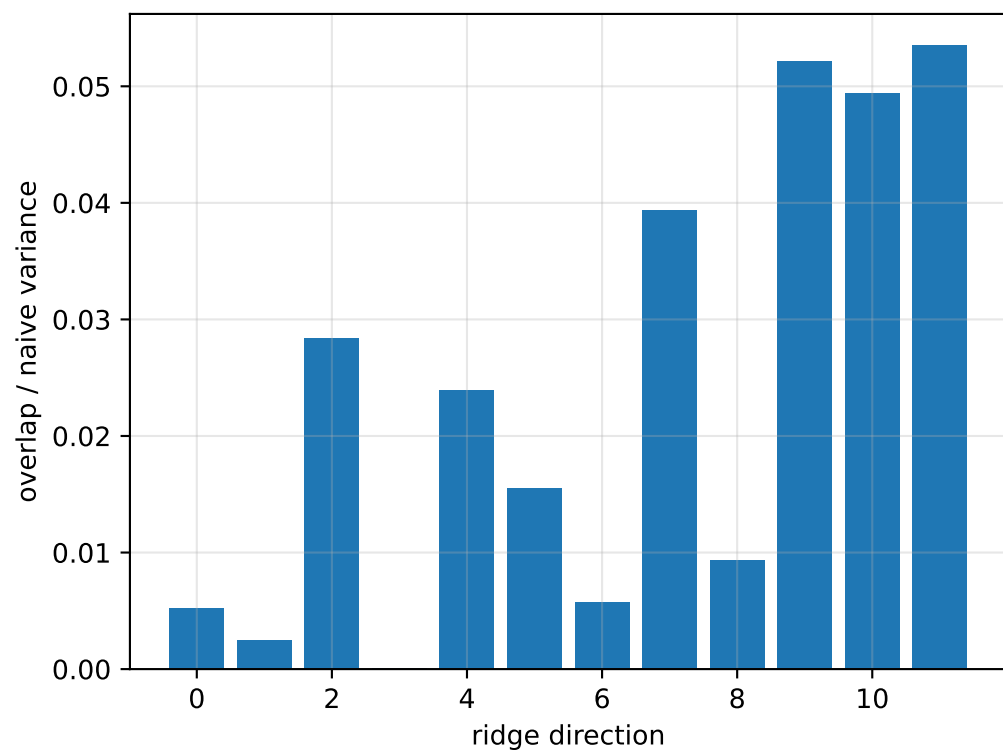
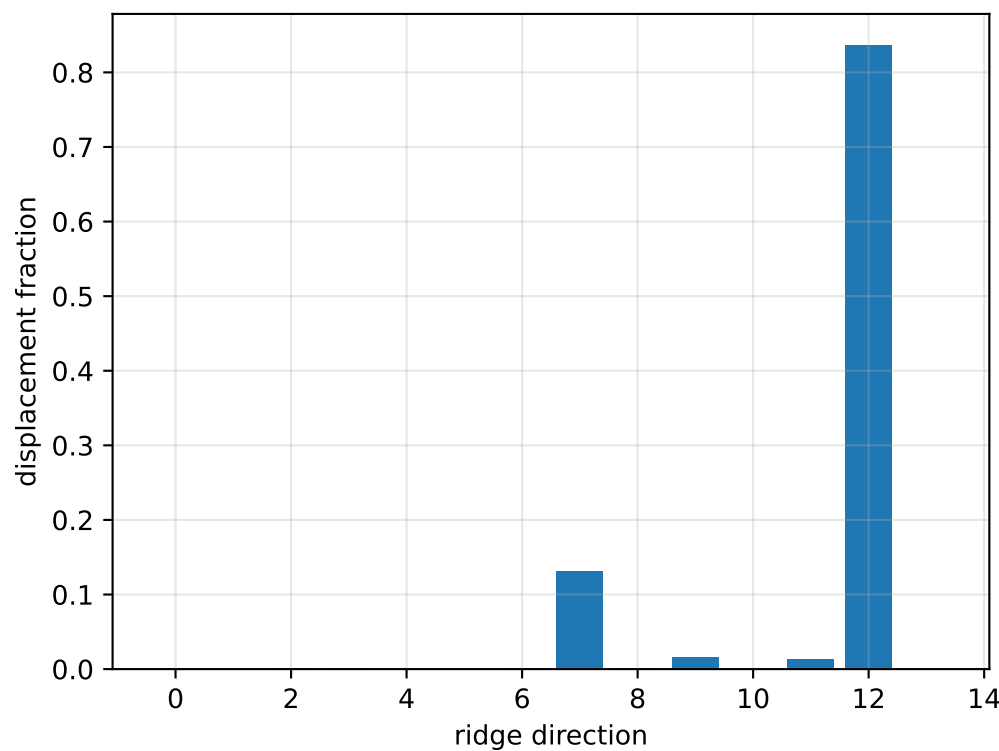
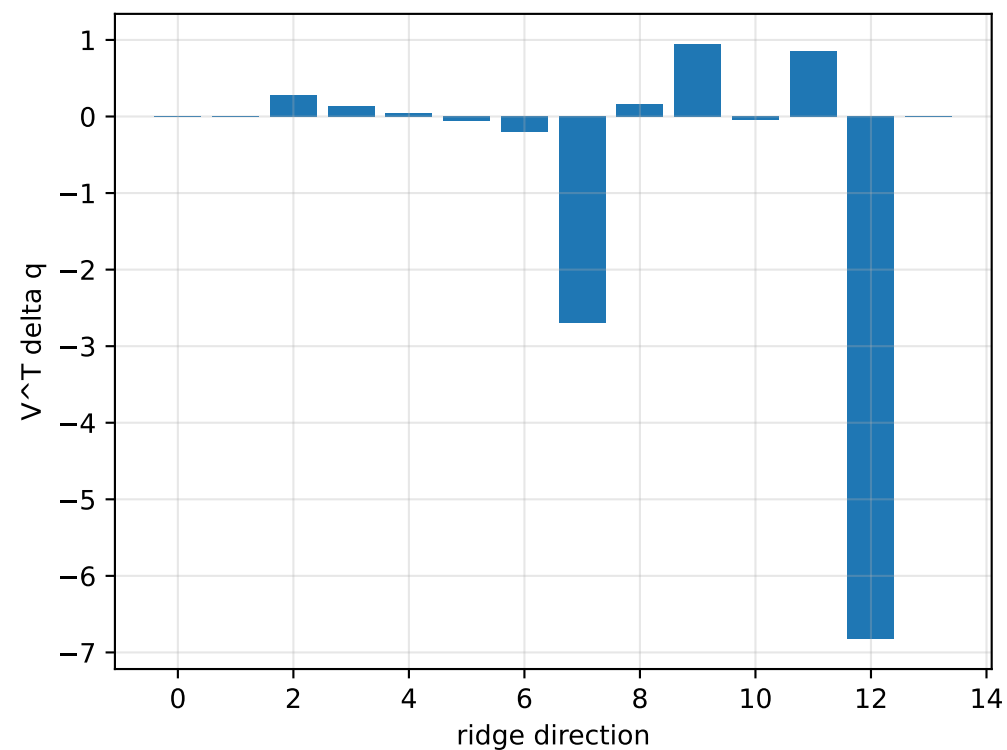
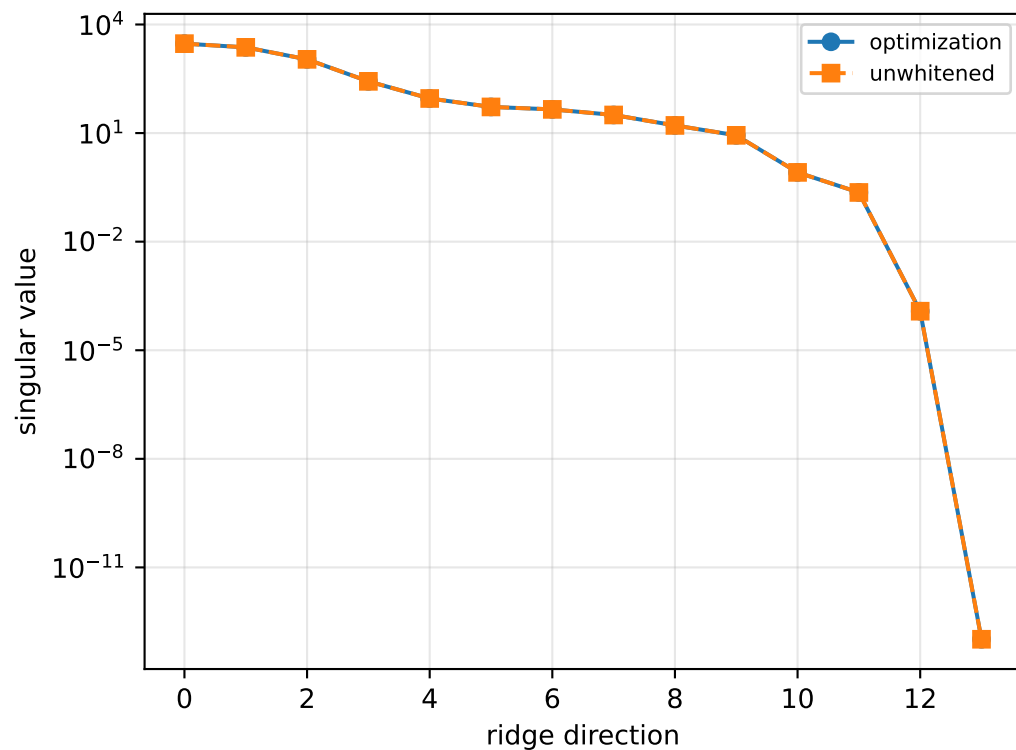
[-1.64915634e-02 9.52257990e-06 1.71547962e-01]]

scale-free f/m: [0.33490906 0.43126309 0.37659816 0.31223847]

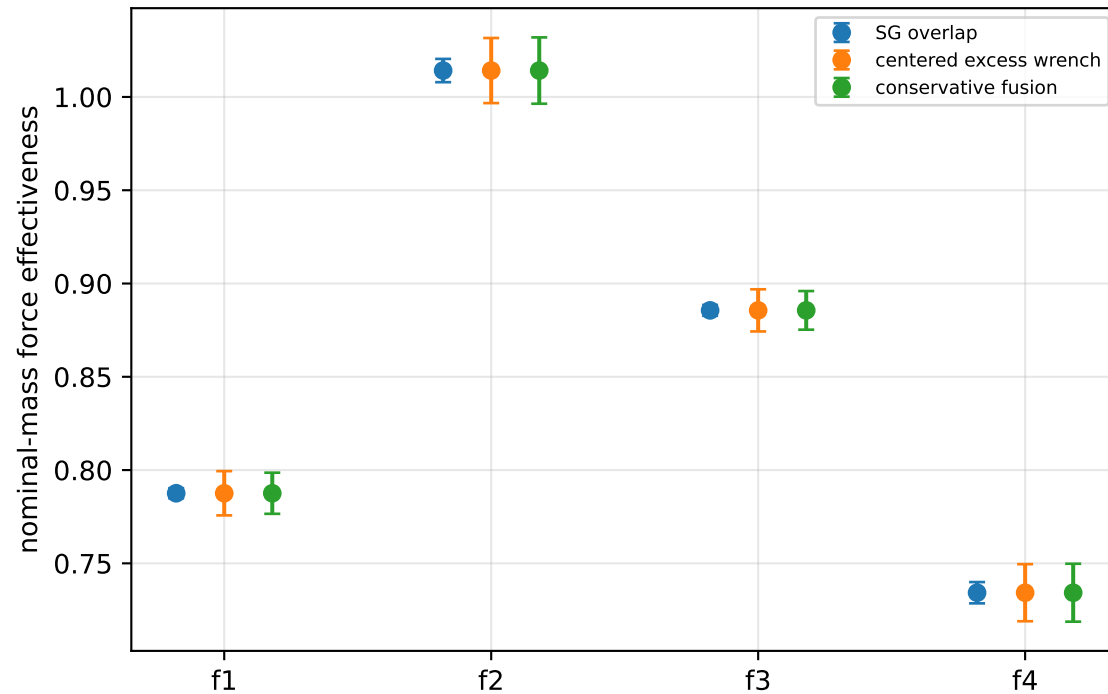
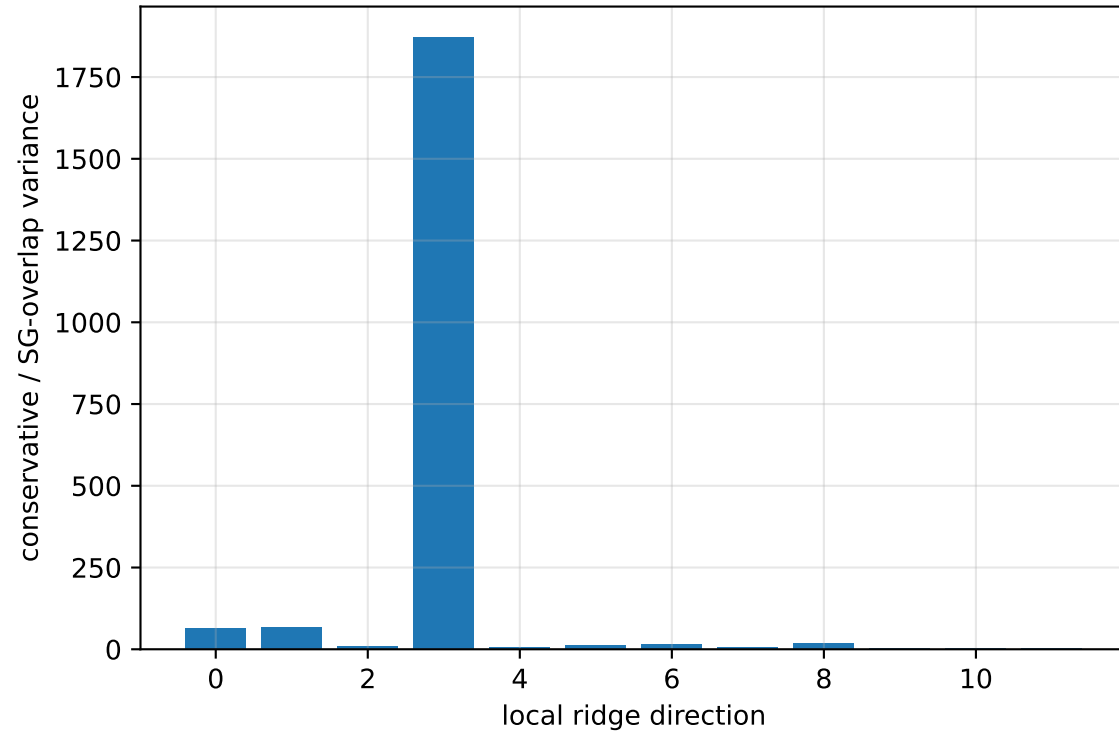
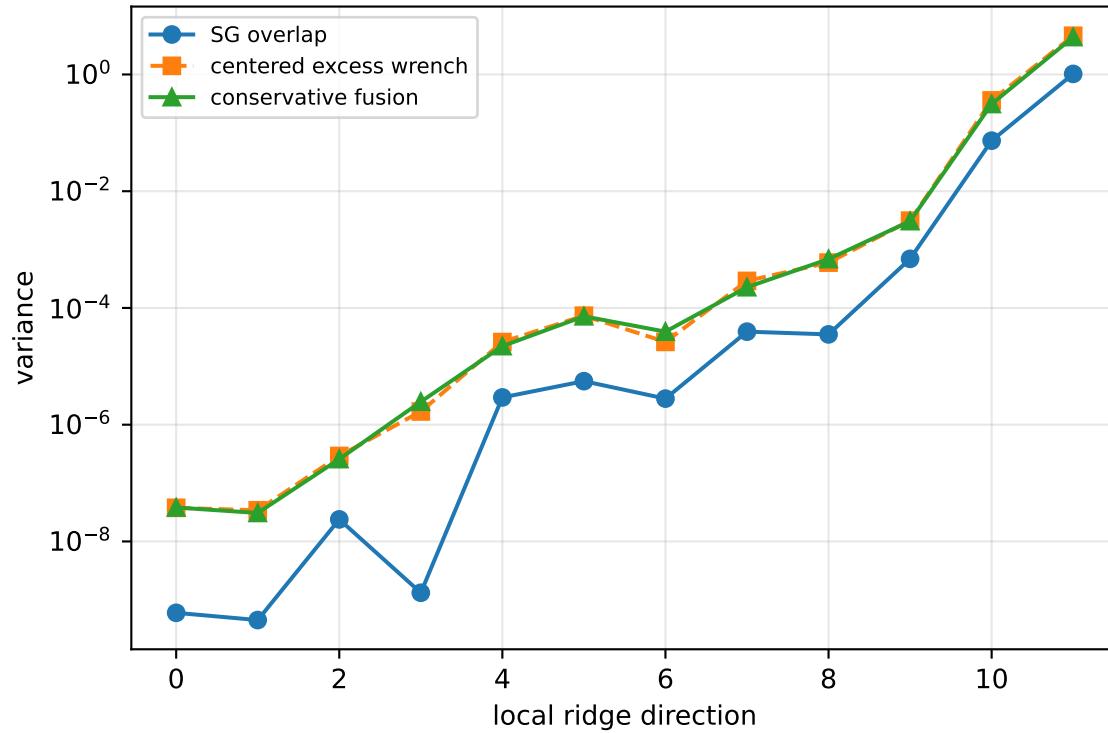
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

default: ridge spectrum (rank 13, nullity 1, $||jv||=1.12e-13$)



default: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.2$
wrench-to-acceleration recovery max error = $3.05\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

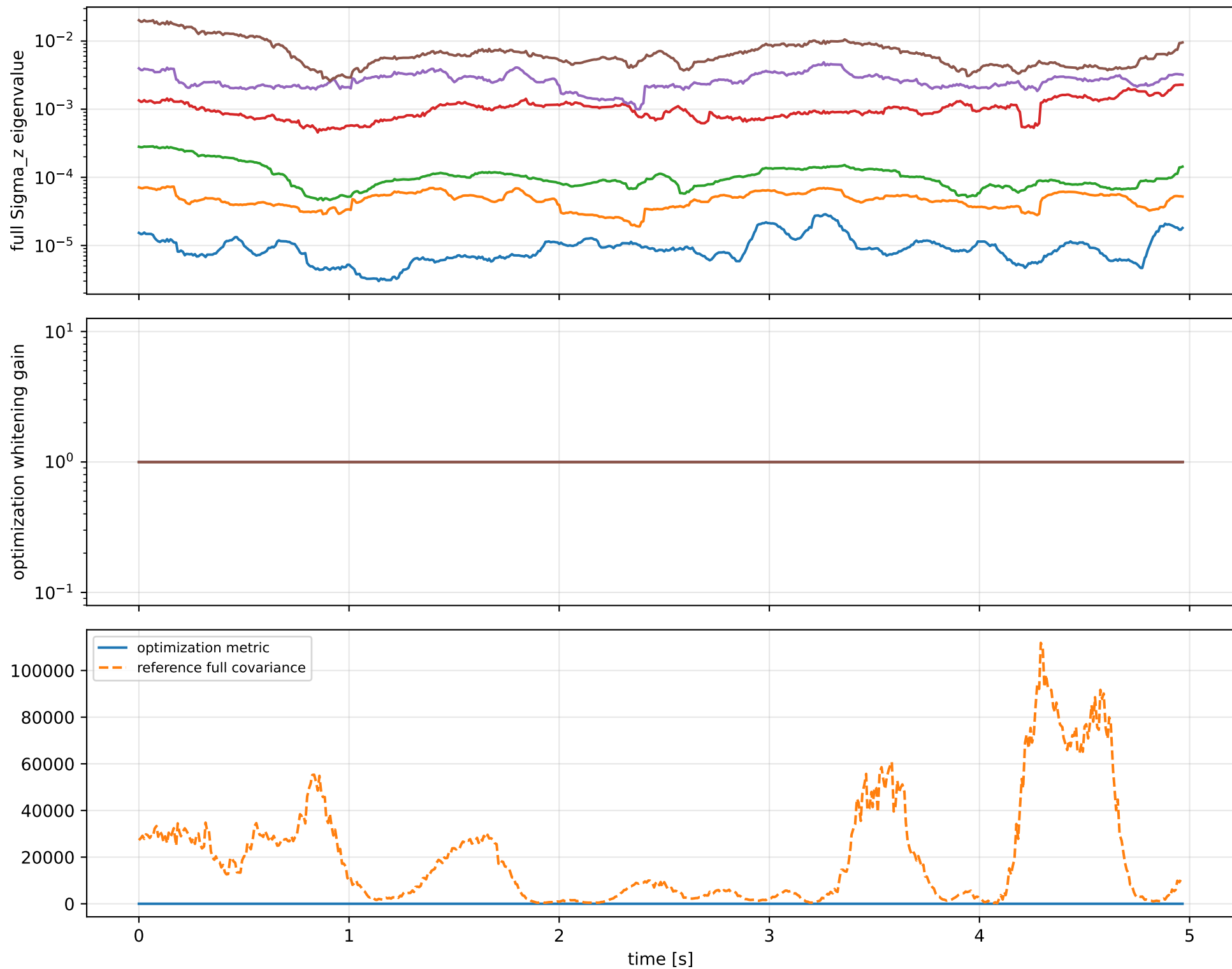
index 11: variance=4.37, ratio=4.275
 $\log_{\text{mass}} = -0.0479$; $\text{second_moment_diag_1} = -0.415$
 $\text{second_moment_diag_2} = -0.0463$; $\text{second_moment_diag_3} = +0.701$
 $\text{second_moment_offdiag_12} = +0.0441$; $\text{second_moment_offdiag_13} = -0.567$
 $\text{second_moment_offdiag_23} = -0.0268$; $\text{cog_x} = +6.41\text{e-}06$
 $\text{cog_y} = -0.000197$; $\text{cog_z} = +2.07\text{e-}05$
 $\log_{\text{force_eff_1}} = -0.0475$; $\log_{\text{force_eff_2}} = -0.0469$
 $\log_{\text{force_eff_3}} = -0.049$; $\log_{\text{force_eff_4}} = -0.0484$

index 10: variance=0.3116, ratio=4.244
 $\log_{\text{mass}} = +0.00245$; $\text{second_moment_diag_1} = +0.0214$
 $\text{second_moment_diag_2} = +0.0107$; $\text{second_moment_diag_3} = -0.0402$
 $\text{second_moment_offdiag_12} = +0.884$; $\text{second_moment_offdiag_13} = +0.0239$
 $\text{second_moment_offdiag_23} = -0.463$; $\text{cog_x} = +0.00142$
 $\text{cog_y} = -0.00354$; $\text{cog_z} = +0.000339$
 $\log_{\text{force_eff_1}} = +0.0105$; $\log_{\text{force_eff_2}} = +0.0204$
 $\log_{\text{force_eff_3}} = -0.00517$; $\log_{\text{force_eff_4}} = -0.0202$

index 9: variance=0.003069, ratio=4.423
 $\log_{\text{mass}} = -0.148$; $\text{second_moment_diag_1} = -0.034$
 $\text{second_moment_diag_2} = +0.899$; $\text{second_moment_diag_3} = -0.0816$
 $\text{second_moment_offdiag_12} = -0.00946$; $\text{second_moment_offdiag_13} = -0.0844$
 $\text{second_moment_offdiag_23} = +0.0136$; $\text{cog_x} = +0.0221$
 $\text{cog_y} = -0.0136$; $\text{cog_z} = -0.00395$
 $\log_{\text{force_eff_1}} = -0.187$; $\log_{\text{force_eff_2}} = -0.0222$
 $\log_{\text{force_eff_3}} = -0.094$; $\log_{\text{force_eff_4}} = -0.332$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

default: optimization vs reference covariance



default: wrench / closure (force max 1.07e-14, torque max 7.64e-16)

